

DTSA 5504

Data Mining Pipeline

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Core	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 3 – Core	Applied Scientist (40.5 / 100)	38.1 / 100
SPECIALIZATION Data Mining Foundations and Practice		

38

CURRENT AVG SCORE
across 6 job roles

79

POTENTIAL SCORE
with recommended additions

+47

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course introduces the key steps involved in the data mining pipeline, including data understanding, data preprocessing, data warehouse, data modeling, interpretation and evaluation, and real-world applications.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Applied Scientist	40.5	87.2	+46.7	HIGH
Data Scientist	39.5	84.5	+45.1	HIGH
Machine Learning Engineer	38.9	80.1	+41.2	HIGH
AI Engineer	36.7	76.5	+39.7	HIGH
MLOps Engineer	36.7	77.9	+41.2	HIGH
ML Engineer	36.0	69.1	+33.1	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

LLMs & Generative AI 2 of 6 roles

storage

Deep Learning 1 of 6 roles

training

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **MEDIUM** **Consider LLMs & Generative AI content** — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: storage.

2 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

3 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Applied Scientist

40.5 → 87.2 (+46.7 pts)

storage literature problems models technical learning industry real

Data Scientist

39.5 → 84.5 (+45.1 pts)

sets analytics different social problems

Machine Learning Engineer

38.9 → 80.1 (+41.2 pts)

web sets learning analytics exploratory business science models

AI Engineer

36.7 → 76.5 (+39.7 pts)

following models learning technical problems storage detection machine

MLOps Engineer

36.7 → 77.9 (+41.2 pts)

detection text learning machine training models

ML Engineer

36.0 → 69.1 (+33.1 pts)

learning machine models world problems technical history detection

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.

